CHAPTER 13

Master Formula Sheet & Mixed Practice

Every formula of Chapters 2–12 in one place · exam traps · 12 mixed questions with answers

QUICK TRICK

Read this sheet the night before. Each row points to its chapter; reopen the chapter if a row looks unfamiliar.

13.1 Formula sheet

● LEVEL 2 · EXAM

Topic (Ch)	Formula / fact
Performance (2)	$T = IC \times CPI / f$ · $CPI = \Sigma\, F_i CPI_i$ · $MIPS = f/(CPI \times 10^6)$ · speedup $= T_{old}/T_{new}$
Amdahl (2)	$S = 1 / [(1 - f) + f/k]$ · $\max = 1/(1 - f)$
Integers (3)	2's complement $-2^{n-1} \ldots 2^{n-1} - 1$ · negate = invert + 1 · $V = C_{in,MSB} \oplus C_{out,MSB}$
IEEE 754 (3)	single 1|8|23 bias 127 · double 1|11|52 bias 1023 · $(-1)^s \times 1.f \times 2^{e-bias}$ · e = 0 denormal/zero · e = max inf/NaN
Instruction (4)	opcode bits = $\lceil \log_2 N \rceil$ · expanding opcode: (left) $\times\, 2^{field}$ · (A+B)(C+D): 3/6/7/8 instructions for 3/2/1/0-address
Addressing (5)	Direct EA = A · Indirect EA = M[A] · Indexed A + R · PC-relative = PC_{next} + offset · memory refs: imm 0, direct 1, indirect 2
ALU (6)	FA: $S = A \oplus B \oplus C$, $C_{out} = AB + C(A \oplus B)$ · CLA $C_{i+1} = G_i + P_i C_i$ · Booth 10 sub / 01 add · product 2n bits
Control (7)	horizontal width = signals (+next address) · vertical $\Sigma \lceil \log_2(n_i + 1) \rceil$ · store bits = words × width
Pipeline (8)	cycles = $k + n - 1$ · speedup = $nk/(k + n - 1) \le k$ · clock = max stage · CPI = 1 + stalls · RAW: 2 / 0 / 1 stalls
Memory perf. (9)	AMAT = $H + m \cdot P$ · 2-level $H_1 + m_1(H_2 + m_2 P)$ · parallel $h \cdot t_c + (1-h) t_m$ · hierarchical + t_c on miss
Interfacing (9)	chips = (locs ratio) × (width ratio) · address lines = $\log_2$(locs) · low-order interleave
Cache (10)	offset = $\log_2 B$ · sets = lines/k · tag = A − index − offset · bits = lines × (data + tag + valid [+ dirty])
Disk (11)	access = seek + ½(60/RPM) + transfer · capacity = surf × tracks × sect × bytes
I/O (12)	overhead = events/s × cycles / f · DMA: 1 interrupt per block · cycle stealing vs burst

13.2 Ten traps that cost marks

COMMON MISTAKE

- Speedup of a pipeline can never exceed k; the clock is the **slowest** stage plus latch.
- Load-use needs a stall (1) even with forwarding.
- Tag bits come from the address width minus index minus offset; fully associative has no index.
- Total cache size must include tag, valid and (write-back) dirty bits.
- PC-relative uses the **incremented** PC.
- 2's complement range is asymmetric; carry out ≠ signed overflow.
- IEEE 754 has a hidden leading 1; bias is 127 (single) / 1023 (double).
- Amdahl uses fractions of the **original** time.
- Simultaneous vs hierarchical access formulas differ: read the question.
- DMA: one interrupt per **block**, but the CPU still sets it up.

13.3 Mixed practice

Practice questions

● LEVEL 2 · EXAM

Q1. A program has 5×10^8 instructions, average CPI 2.2, on a 1.1 GHz CPU. Execution time?

Q2. Encode −0.15625 in IEEE single precision (hex).

Q3. Addressing: M[100] = 200, M[200] = 300, R1 = 100. What does "LOAD (100)" (indirect) give? And indexed 100(R1)?

Q4. A 5-stage pipeline executes 200 instructions with no stalls. Cycles and speedup over a non-pipelined machine (5 cycles each)?

Q5. Chips needed for a 32K × 16 memory from 8K × 8 chips.

● LEVEL 3 · GATE

Q6. 32-bit address, 8 KB 2-way cache, 16 B blocks. Index and tag bits?

Q7. Instruction mix 40% ALU (1 cycle), 30% load (2 cycles + 1 stall per 4 loads on average), 30% branch (2 cycles). Average CPI?

Q8. LW R1,0(R2); ADD R3,R1,R4 on a pipeline with forwarding. Stall cycles?

Q9. Microprogrammed control store: 1024 words, 36 control signals (horizontal) plus a next-address field. Width and size?

Q10. Cache: 95% hit, hit time 2 ns, memory 100 ns (hierarchical access). Average access time?

Q11. Disk: 7200 RPM, average seek 8 ms, 4 KB read with 0.5 ms transfer. Access time?

Q12. A device sends 1 MB/s in 4-byte words using cycle-stealing DMA with a 200 ns memory cycle. CPU time lost?

Answers

Q	Solution
1	$5\times10^8 \times 2.2 / 1.1\times10^9$ = **1 s**.
2	**0xBE200000** (1.01×2^{-3}, exponent 124).
3	Indirect: M[M[100]] = **300**. Indexed: EA = 200 → **300**.
4	5 + 199 = **204 cycles**. Speedup = 1000/204 ≈ **4.9**.
5	(32/8) × (16/8) = 4 × 2 = **8**.
6	Lines = 512, sets = 256 → index **8**; offset 4; tag = 32 − 12 = **20**.
7	Load effective = 2 + 0.25 = 2.25. CPI = 0.4 × 1 + 0.3 × 2.25 + 0.3 × 2 = 0.4 + 0.675 + 0.6 = **1.675**.
8	**1** (load-use).
9	Address 10 bits → width 36 + 10 = 46 bits. Size = 1024 × 46 = **47,104 bits** (5.75 KB).
10	0.95 × 2 + 0.05 × (2 + 100) = 1.9 + 5.1 = **7.0 ns**.
11	8 + 4.17 + 0.5 = **12.67 ms**.
12	Words/s = 250,000; 250,000 × 200 ns = 50 ms/s = **5%**.

Done. Re-attempt the mixed practice after two days, then solve previous-year GATE questions chapter by chapter.